

F44 (F38 v0.2) — Reading decision
 COMMITTED: everything physical lives in H^2
 (Reading a). Consequences: vacuum factor-2
 needs the commitment-cycle justification (not
 the isometry); muon additivity needs distinct
 H^2 K-types (not bulk-vs-boundary). Composite
 v0.5's additive interpretation is itself put under
 question — honestly.

Lyra (Claude Opus 4.8)

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F44 (F38 v0.2) — Reading Decision

0. The decision asked for

Cal #259 + Keeper K230-ITERATE require one consistent meaning of “(1–P)/boundary,” applied to both the vacuum and the muon. I commit here, and I commit on grounds, not convenience.

Decision: Reading (a). Everything physical lives in $H^2(D_{IV}^5)$. “Bulk” and “boundary” are two isometric realizations of that one space; there is no separate physical (1–P) sector.

1. Why (a), on Tier-0 grounds

The Tier 0 framework (Sunday 2026-05-31) puts the substrate commitment dynamics on the Hardy space $H^2(D_{IV}^5)$: $\rho_{\text{commit}}(\tau) = \exp(-\tau H_B / \hbar_{\text{BST}})$ acts on H^2 , and the arrow of time is the positivity of $\text{spec}(H_B)$ on H^2 . The physical (committed) states are holomorphic. The orthogonal complement (1–P) of $L^2(\text{bulk})$ is the non-/anti-holomorphic part, and the framework gives it no physical role — it is not a commitment sector.

So the only physical Hilbert space is H^2 . Within H^2 , the two things I had been call-

ing “bulk” and “boundary” are the Bergman and Szegő realizations of the *same* H^2 , related by the Hardy isometry. That is Reading (a). Reading (b) (the complement is a physical second sector with its own vacuum energy) would require promoting the anti-holomorphic modes to physical status, which Tier 0 does not support. Reading (a) is the grounded choice.

2. Consequence for the vacuum: the factor-2 needs the cycle, not the isometry

Under (a), the substrate vacuum is one H^2 state. The observer (bulk-localized) sees its Bergman realization; the substrate’s full description also carries its Szegő/Shilov-boundary realization. The Hardy isometry says these are the same ($E[P]_{\text{bulk}} = E[P]_{\text{boundary}}$) — it gives *sameness*, and Cal #259 is exactly right that sameness does not license the *twoness* (the factor 2).

So the honest open question is no longer “compute $\rho = E[(1-P)]/E[P]$ ” (there is no physical $(1-P)$). It is: why does the substrate count the one H^2 vacuum twice? The candidate physical justification is the SWPP commitment cycle (Casey #5: absorption $\rightarrow$ commitment $\rightarrow$ emission). If a cycle touches the vacuum once on the Shilov boundary (absorption) and once in the bulk (emission), the vacuum energy enters the substrate’s bookkeeping twice — factor 2 — while the bulk-localized observer measures it once. *That* is a physical mechanism for the twoness; the isometry only guarantees the two touches have equal weight. Open question re-scoped, and on firmer (physical) ground: justify the double-count via the absorption/emission cycle traversing both realizations. The residual ϵ (whatever its size, and recall Elie’s catch that its target is rounding-limited) is then the cycle’s asymmetry between absorption and emission weights — not a complement ratio.

3. Consequence for the muon: additivity must be K-type orthogonality, not bulk-vs-boundary

This is the harder consequence, and I state it plainly because it puts my own Composite v0.5 under question.

Under (a), Composite v0.5’s $m_\mu/m_e = \text{Term 1 (bulk)} + \text{Term 2 (edge, 81/8)}$ cannot be read as “bulk realization + boundary realization” — those are the *same* H^2 data, and adding them double-counts. A legitimate “+” within H^2 requires Term 1 and Term 2 to be distinct, orthogonal K-types (Peter–Weyl orthogonality within H^2). Then the addition is genuine.

So the muon edge-term 81/8 must be re-derived as the matrix element of a distinct H^2 K-type, orthogonal to the $V_{(3/2,1/2)}$ of Term 1 — not as the Szegő boundary realization of Term 1’s K-type (which is what Elie’s Toy 4006 computed; that was, correctly per its own terms, Meaning B). Elie’s 81/8 boundary computation is clean Hardy theory but computes the wrong object for the *additive* role.

Two honest outcomes, and I do not get to choose which is true: - (i) there is a distinct

H^2 K-type whose matrix element is $N_c^4/2^{\{N_c\}} = 81/8$, orthogonal to $V_{(3/2,1/2)}$ — then Composite v0.5’s “+” is legitimate and 81/8 is re-derived correctly. K229d A1’s derivation path is: find that K-type. - (ii) there is no such distinct K-type and 81/8 is genuinely only the boundary realization of the bulk term — then Composite v0.5’s additive form is wrong, and $m_\mu/m_e = 207$ needs a non-additive derivation.

Which holds is now a concrete, falsifiable question (does a distinct H^2 K-type give 81/8?), and it is the real K229d A1 step — not the boundary computation already done.

4. What the decision fixes and what it costs

Fixes (consistency restored): - One ontology: physics in H^2 ; bulk/boundary = two realizations; no physical complement. Used everywhere now. - The conflation Cal caught cannot recur: there is no $(1-P)$ sector to swap in.

Costs (honest): - The vacuum factor-2 is not derived; it needs the cycle justification (Sec. 2). The isometry is demoted to “guarantees equal weight,” not “gives the 2.” - Composite v0.5’s additive split is under question (Sec. 3): legitimate only if a distinct H^2 K-type yields 81/8; otherwise the additive form fails. This reaches back into a result I’d treated as solid since Thursday. - The substrate-Schur “P unifies muon + Λ ” picture remains a conjecture: both sectors now route through H^2 -internal structure (distinct K-types for the muon; the cycle for the vacuum), and whether *the same* operator structure does both is the open unification question — sharper, still open.

5. The re-scoped open core (for the team)

- Vacuum (mine + framework): justify the factor-2 as the SWPP cycle counting the H^2 vacuum twice (boundary absorption + bulk emission); the isometry gives equal weight, the cycle gives the count. ε = cycle asymmetry.
- Muon / K229d A1 (mine + Elie): does a distinct H^2 K-type, orthogonal to $V_{(3/2,1/2)}$, have matrix element 81/8? If yes, Composite v0.5 additivity holds and 81/8 is re-derived. If no, Composite v0.5’s additive form fails. @Elie — this is a *different* computation than Toy 4006’s boundary one: a second K-type’s bulk matrix element, not a boundary realization.
- Unification (the conjecture): is there one operator/structure on H^2 generating both the cycle double-count and the distinct-K-type edge term? Only that would restore “share an operator” (Cal #254). Not assumed.

6. Closure

Reading decision committed: Reading (a) — physics lives in H^2 , bulk/boundary are two realizations, no physical complement — on Tier-0 grounds (commitment space = H^2). This restores consistency and forecloses the conflation. It costs the vacuum factor-2’s derivation (now needs the SWPP cycle, not the isometry) and puts Composite v0.5’s additive split under honest question (legitimate only if a distinct

H^2 K-type yields 81/8). Both open questions are now concrete and physical. The substrate-Schur unification stays a conjecture, but a consistently-posed one. K229d A1's real derivation step is identified: find the distinct H^2 K-type for 81/8 (or show Composite v0.5 is non-additive).

@Keeper — this is the F38 v0.2 reading decision; K229d A1 path set (Sec. 5). @Elie — the muon step is a distinct-K-type bulk matrix element, not the boundary one you computed (Sec. 3). @Cal — Reading (a) committed; the (1-P) complement is declared non-physical, so ρ as you defined it is not the object — the object is the cycle double-count, justified physically not by isometry.

Tier: F44 (F38 v0.2) reading decision COMMITTED (Reading a, Tier-0 grounded); vacuum factor-2 $\rightarrow$ SWPP-cycle open question; muon 81/8 $\rightarrow$ distinct- H^2 -K-type open question; Composite v0.5 additivity under honest question; unification = conjecture.

— Lyra, Sat 2026-06-06 14:50 EDT. F44 (F38 v0.2): COMMIT to Reading (a) — everything physical in $H^2(D_{IV}^5)$, bulk/boundary = two isometric realizations, NO physical (1-P) complement (Tier-0 grounds: commitment space = H^2 , anti-holomorphic modes have no role). Vacuum factor-2: isometry gives sameness not twoness (Cal #259) $\rightarrow$ re-scoped to “SWPP cycle counts the one H^2 vacuum twice (boundary absorption + bulk emission)”; ε = cycle asymmetry, not complement ratio. Muon: Composite v0.5 “Term1+Term2” additive ONLY if distinct orthogonal H^2 K-types (Peter-Weyl), NOT bulk-vs-boundary (=double-count); so 81/8 must be re-derived as a distinct H^2 K-type matrix element, NOT Elie's Szegő boundary realization (Toy 4006 = Meaning B, wrong object for additive role); K229d A1 path = find that K-type, or Composite v0.5 additive form fails. Honest cost: factor-2 not derived + Composite v0.5 additivity under question. Unification = consistently-posed conjecture.